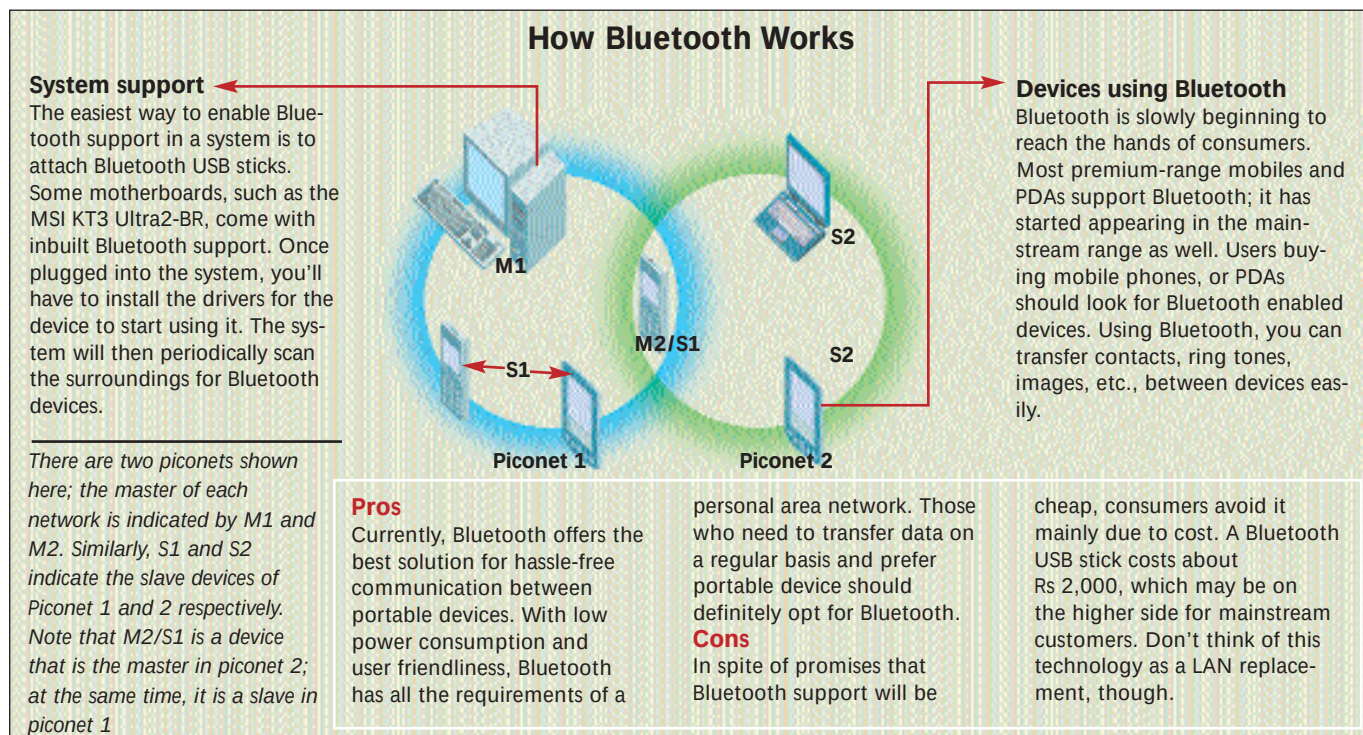




hopping. The devices frequently change the frequency at which they operate—as often as 1,600 times a second. This prevents interference with other devices that may be operating at the same frequency. When a piconet is formed, the master informs all the slaves of the frequency hopping sequence, and the slaves follow these instructions.

To conserve power, devices can go into three modes when they aren't actively involved in a piconet. In increasing order of power consumption, they can be in park mode, hold mode and sniff mode. All the slave devices in a piconet are assigned an active member address (AM_ADDR). When a slave enters the park mode, it will give up its AM_ADDR and get a park address, PM_ADDR. The parked devices will listen to the master at regular intervals, allowing them to conserve power, and will yet be able to rejoin a network when required. When a device enters hold mode, it is assigned a hold timer and no data transfer takes place until the timer ends. In sniff mode, the device retains its AM_ADDR, but reduces its role in the piconet, thus conserving power. This mode is flexible, and parameters such as sniffing interval can be programmed.

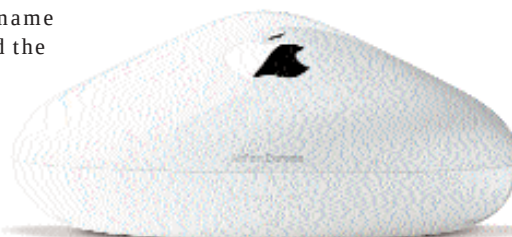
One of Bluetooth's advantages is that it can handle both data (asynchronous) and voice (synchronous), which others such as infra-red can't. One must under-

stand that the requirements of a voice and data channel are very different. A data channel needs to be strictly error free while it can tolerate timing errors, for a voice channel it is exactly the opposite. Both synchronous and asynchronous channels can be handled simultaneously. This has led to products such as Bluetooth headsets for mobile phones; these headsets allow users to talk via the cordless headsets while the mobile phone resides in their pockets.

Every Bluetooth device has a 48-bit BDA (Bluetooth Device Address) burned into its ROM. This address can't be easily changed by the user. Similarly, each device can be given a user-friendly name such as 'Sachin's mobile'. These names will make it easy for the user to identify a device in a piconet. A Bluetooth device can be set to periodically scan for other devices in its vicinity, or users can perform manual scans. After a scan is performed, a list of all Bluetooth devices detected is shown. Each device in the list will have the name assigned by its respective owner, and the user can select a device and communicate with it. A Bluetooth device will show up in a scan only when it is in discoverable mode. Discoverable mode should be turned off to prevent bluejacking—sending anonymous messages to users via Bluetooth.

WiFi

Bluetooth is a convenient way of communication for a personal network, but it is not suitable as an LAN replacement, or for perennial data transfer. But the growing popularity of laptops necessitated the need for a wireless standard that would allow people to walk around with their laptops, and still be connected to a network. WiFi, an abbreviation for Wireless Fidelity, is a wireless standard that promises mobility while offering data rates comparable to those of a wired LAN. WiFi is a collection of standards ratified by the IEEE (Institute of Electrical and Electronics Engineers). In 1997, IEEE approved the 802.11 standard, which laid down the specifications for wireless LAN. Subsequently, revisions were made to this standard and these resulted in three other standards, namely 802.11a, 802.11b and 802.11g. The entire family of 802.11 standards are collectively called WiFi. The letters appended to the standard



The Apple Airport can be plugged into a Mac system to make it WiFi compatible